

GRANADA AP, Spain

WMO: 084190

Lat: 37.190N

Lon: 3.789W

Elev: 567

StdP: 94.70

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.5	-2.1	-8.0	2.0	3.7	-5.8	2.5	3.4	9.2	9.9	7.8	10.0	0.5	210	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	38.2	20.2	36.9	19.8	35.2	19.3	22.5	34.2	21.3	32.8	20.4	31.8	4.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	14.3	28.2	17.4	13.3	26.4	16.4	12.5	25.3	68.9	34.4	64.3	32.7	60.8	31.7	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.1	7.1	6.2	-5.6	40.7	2.1	1.6	-7.2	41.9	-8.4	42.8	-9.6	43.7	-11.2	44.9		
			-6.2	23.9	2.2	2.4	-7.8	25.6	-9.1	27.0	-10.3	28.4	-11.9	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	7.2	8.6	11.5	14.0	18.1	23.0	25.9	25.6	21.2	17.0	10.8	8.0	(6)
(7)		DBStd	7.21	2.58	2.82	2.79	3.07	3.51	3.12	2.38	2.41	2.77	3.02	3.28	2.73	(7)
(8)		HDD10.0	273	93	55	18	4	0	0	0	0	1	29	74		(8)
(9)		HDD18.3	1631	346	274	213	132	48	4	0	6	62	225	322		(9)
(10)		CDD10.0	2446	5	14	63	126	252	391	493	484	337	216	54	11	(10)
(11)		CDD18.3	761	0	0	0	3	41	145	235	226	93	19	0	0	(11)
(12)		CDH23.3	11016	0	1	31	154	734	2102	3426	3072	1179	310	8	0	(12)
(13)		CDH26.7	5867	0	0	3	30	282	1109	2064	1783	514	81	0	0	(13)
(14)	Wind	WSAvg	1.9	1.4	1.9	2.1	2.3	2.3	2.4	2.3	2.1	1.8	1.5	1.5	1.3	(14)
(15)	Precipitation	PrecAvg	382	42	40	48	39	30	6	0	2	27	44	57	46	(15)
(16)		PrecMax	564	159	126	191	93	69	39	0	14	65	107	159	184	(16)
(17)		PrecMin	185	0	0	5	1	4	0	0	0	16	7	1		(17)
(18)		PrecStd	108	38	30	49	21	18	10	0	4	22	26	36	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.2	22.2	26.2	29.7	34.5	39.0	40.9	39.9	36.2	31.3	24.6	19.8	(19)
(20)		0.4%	MCWB	10.9	12.5	14.1	15.4	18.0	19.2	22.0	20.4	18.5	16.2	14.4	11.6	(20)
(21)		2%	DB	16.9	19.8	23.9	27.2	31.9	36.4	38.6	38.0	33.5	29.0	21.9	17.3	(21)
(22)		2%	MCWB	10.1	11.3	13.2	14.8	17.2	19.2	20.5	20.1	18.6	16.3	13.5	11.1	(22)
(23)		5%	DB	15.0	17.6	21.8	25.0	29.9	34.6	37.1	36.3	31.6	26.8	19.7	15.2	(23)
(24)		5%	MCWB	9.3	10.4	12.3	14.1	16.4	18.7	20.0	19.8	17.9	15.9	12.9	10.7	(24)
(25)		10%	DB	13.1	15.3	19.4	22.6	27.8	32.8	35.6	34.8	29.8	24.7	17.5	13.8	(25)
(26)		10%	MCWB	8.9	9.7	11.4	13.3	15.6	18.2	19.3	19.5	17.5	15.6	12.2	10.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	13.5	15.1	17.0	19.8	22.5	25.0	24.1	21.2	19.0	16.6	13.7	(27)
(28)		0.4%	MCDWB	16.2	19.6	23.9	26.5	30.8	33.8	37.6	36.0	31.1	26.3	20.6	16.2	(28)
(29)		2%	WB	11.4	12.2	13.7	15.6	18.2	20.8	22.7	22.4	20.1	17.9	14.8	12.5	(29)
(30)		2%	MCDWB	14.7	17.5	21.7	25.2	29.3	32.6	35.2	33.6	29.6	24.9	19.5	15.5	(30)
(31)		5%	WB	10.3	11.2	12.7	14.7	17.2	19.7	21.2	21.4	19.3	16.9	13.8	11.4	(31)
(32)		5%	MCDWB	13.6	16.1	20.1	23.6	27.7	31.8	33.6	32.4	28.5	23.8	18.0	14.6	(32)
(33)		10%	WB	9.2	10.2	11.8	13.7	16.2	18.9	20.1	20.5	18.5	16.2	12.7	10.3	(33)
(34)		10%	MCDWB	12.5	14.7	18.4	21.6	26.0	30.9	32.7	31.5	27.2	22.8	16.6	13.3	(34)
(35)	Mean Daily Temperature Range		MDBR	12.7	13.4	14.3	14.2	15.3	17.4	18.8	18.2	15.9	14.5	12.7	12.4	(35)
(36)		5% DB	MCDBR	16.6	18.0	19.6	19.5	19.9	20.1	20.9	20.1	19.2	18.4	16.4	15.2	(36)
(37)			MCWBR	10.2	10.4	10.3	9.1	8.2	7.4	8.0	7.2	7.3	7.8	8.9	9.6	(37)
(38)			MCDBR	12.9	14.9	17.2	17.6	17.9	18.6	19.6	18.1	16.9	15.1	12.7	12.2	(38)
(39)		5% WB	MCWBR	8.4	9.0	9.3	8.6	7.9	8.0	9.0	8.0	7.0	7.4	7.4	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.326	0.366	0.364	0.381	0.400	0.391	0.398	0.377	0.361	0.324	0.304	(40)	
(41)		taud	2.517	2.452	2.333	2.367	2.349	2.309	2.330	2.315	2.395	2.431	2.505	2.522	(41)	
(42)		Ebn at Noon	898	908	896	916	901	880	885	872	874	855	857	865	(42)	
(43)		Edh at Noon	78	95	117	120	124	129	126	125	109	96	79	74	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.62	3.42	4.66	5.77	6.86	7.81	7.89	6.91	5.37	3.90	2.83	2.34	(44)	
(45)		RadStd	0.28	0.42	0.48	0.47	0.53	0.22	0.17	0.21	0.32	0.31	0.27	0.22	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.78	N/A	N/A	N/A	N/A	N/A	+85
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+76	+62

Nomenclature: See separate page